



HWA CHONG INSTITUTION
C2 Preliminary Examination
Higher 2

NAME

CT GROUP

21S

CHEMISTRY

9729/04

Paper 4 Practical

25 August 2022

2 hours 30 minutes

Candidates answer on the Question Paper

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Give details of the practical shift and laboratory where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.
Qualitative Analysis Notes are printed on pages 16 and 17.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	
2	
3	
4	
Total	

1 Determination of the basicity of an acid and the strength of a base

FA 1 is 1.50 mol dm^{-3} sodium hydroxide, NaOH.

FA 2 is 0.75 mol dm^{-3} acid **A**, which is either monobasic (monoprotic) or dibasic (diprotic).

FA 3 is 1.50 mol dm^{-3} base **B**, which is monoacidic (monoprotic) and is either a strong base or a weak base.

You will perform experiments to determine whether acid **A** is monobasic or dibasic. You will then calculate the enthalpy change of neutralisation, ΔH_{neut} , between NaOH and acid **A**.

You will also design one more experiment to determine whether base **B** is a strong base or a weak base.

(a) (i) Experiment 1

1. Place one polystyrene cup inside a second polystyrene cup. Place these in a glass beaker to prevent them from tipping over.
2. Use a measuring cylinder to transfer 40.0 cm^3 of **FA 1** into the first polystyrene cup. Stir the **FA 1** in the polystyrene cup with the thermometer with **0.2 °C divisions**. Read and record the initial temperature of **FA 1** in Table 1.1.
3. Use another measuring cylinder to measure 20.0 cm^3 of **FA 2**. Add **FA 2** to **FA 1** in the polystyrene cup. Stir the mixture using the same thermometer and record the maximum temperature reached.
4. Wash and dry the thermometer and the polystyrene cup.

Experiment 2

Repeat experiment 1 using 20.0 cm^3 of **FA 1** and 40.0 cm^3 of **FA 2**.

Use your results to determine the maximum change in temperature, ΔT , for each experiment and complete Table 1.1. You may assume that the initial temperature of **FA 2** is the same as that of **FA 1**.

Table 1.1

experiment	Volume of FA 1 / cm^3	Volume of FA 2 / cm^3	Initial temperature of FA 1 / $^{\circ}\text{C}$	Maximum temperature / $^{\circ}\text{C}$	Maximum change in temperature, ΔT / $^{\circ}\text{C}$
1	40.0	20.0			
2	20.0	40.0			

[2]

- (ii) Use your ΔT results of experiments 1 and 2 to deduce whether acid **A** is monobasic or dibasic. Explain your deduction.

.....

.....

.....

.....

.....

.....

.....

..... [2]

- (iii) Hence, calculate the enthalpy change of neutralisation, ΔH_{neut} , between NaOH and acid **A**.

Assume that the specific heat capacity of the reaction mixture is $4.18 \text{ J g}^{-1} \text{ K}^{-1}$, and that the density of the reaction mixture is 1.00 g cm^{-3} .

$$\Delta H_{\text{neut}} = \dots\dots\dots [2]$$

- (iv) A student repeated **experiment 2** but using solutions of double the concentrations (3.00 mol dm^{-3} NaOH and 1.50 mol dm^{-3} acid **A**).

Suggest the effect this would have on the value of ΔT for experiment 2. Hence, deduce and explain the effect this would have on the value for ΔH_{neut} .

.....

.....

..... [2]

(b) (i) Experiment 3

FA 3 contains a monoacidic base, base **B**, that has the same concentration as the NaOH in **FA 1**.

Design one more experiment, the results of which, when compared against **your experiment 2**, will allow you to deduce whether base **B** is a strong or weak base.

Fill Table 1.2 with your chosen volumes of **FA 3** and **FA 2**.

Carry out this experiment.

Complete Table 1.2.

Table 1.2

experiment	Volume of FA 3 /cm ³	Volume of FA 2 /cm ³	Initial temperature of FA 3 /°C	Maximum temperature /°C	Maximum change in temperature, ΔT /°C
3					

[1]

- (ii)** Use your ΔT results of experiment 3 and **experiment 2** to deduce whether base **B** is a strong base or a weak base. Explain your deduction.

.....

.....

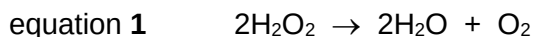
.....

..... [1]

[Total: 10]

2 Investigation of the kinetics of the catalysed decomposition of hydrogen peroxide

Hydrogen peroxide decomposes very slowly to form water and oxygen as shown in equation 1.



Many transition element ions are able to catalyse the decomposition of hydrogen peroxide. Iron(III) nitrate, $\text{Fe}(\text{NO}_3)_3$, is an effective catalyst for this reaction.

FA 4 is iron(III) nitrate, $\text{Fe}(\text{NO}_3)_3$.

FA 5 is $0.0200 \text{ mol dm}^{-3}$ potassium manganate(VII), KMnO_4 .

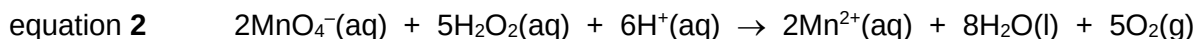
FA 6 is aqueous hydrogen peroxide, H_2O_2 .

FA 7 is 0.2 mol dm^{-3} sulfuric acid, H_2SO_4 . (**FA 7** is also used in **Question 3**.)

You will add a measured volume of **FA 4** to a measured volume of **FA 6** and, at timed intervals, you will transfer aliquots (portions) of the reaction mixture.

It is necessary that you titrate each aliquot against **FA 5** **before** transferring the next aliquot.

Acidified KMnO_4 and H_2O_2 react as shown in equation 2.



(a) (i) Preparation and titration of the reaction mixture

Notes: You will perform each titration **once** only. Great care must be taken that you do not overshoot the end-point.

Once you have started the stopwatch, it must continue running for the duration of the experiment. You must **not** stop it until you have finished this experiment.

You should transfer your first aliquot within the first three minutes of starting the reaction.

The titre volume for your first titration is expected to be greater than 30 cm^3 .

You should **not** exceed a maximum reaction time of 25 minutes for this experiment.

In an appropriate format in the space provided on the next page, prepare a table in which to record for each aliquot

- the time of transfer, t , in minutes and seconds,
- the decimal time, t_d , in minutes, to 0.1 min, for example, if $t = 4 \text{ min } 33 \text{ s}$ then $t_d = 4 \text{ min} + 33/60 \text{ min} = 4.6 \text{ min}$,
- the burette readings and the volume of **FA 5** added.

1. Fill a burette with **FA 5**.
2. Using a measuring cylinder, add 100.0 cm³ of **FA 6** to the conical flask labelled **reaction mixture**.

Do **not** use the H₂O₂ bench reagent for this step.

3. Using a measuring cylinder, add 2.0 cm³ of **FA 4** to the same conical flask. Start the stopwatch and swirl the mixture thoroughly to mix its contents.
4. Using a measuring cylinder, add 50.0 cm³ of **FA 7** to a second conical flask.
5. Before 3 minutes, withdraw a 10.0 cm³ aliquot (portion) of the reaction mixture to a 10 cm³ measuring cylinder, using a dropping pipette.
6. **Immediately** transfer this aliquot into the second conical flask from step 4 and vigorously swirl the mixture. Read and record the time of transfer in minutes and seconds, to the nearest second, when the aliquot is added.
7. **Immediately** titrate the H₂O₂ in the second conical flask with **FA 5**. The end-point is reached when a permanent **pale** pink colour is obtained. Record your titration results.
8. Wash out the second conical flask with water.
9. Repeat steps **4** to **8** until a total of **five** aliquots have been titrated and their results recorded.

You should **not** exceed a maximum reaction time of 25 minutes for this experiment.

Results

[5]

- (a) (ii) Plot a graph of the volume of **FA 5** added, on the y-axis, against decimal time, t_d , on the x-axis on the grid in Fig. 2.1.

Draw the most appropriate best-fit curve taking into account all of your plotted points.

Extrapolate (extend) this curve to the y-axis and read the value of V_0 , the volume of **FA 5** required at $t_d = 0.0$ min.



Fig 2.1

$V_0 = \dots\dots\dots \text{cm}^3$ [4]

(b) The **initial rate** of change of the concentration of hydrogen peroxide, $[\text{H}_2\text{O}_2]$, can be determined from the gradient of the tangent to the graph in Fig. 2.1 at time $t_d = 0.0$ min.

- (i) Draw a tangent to your graph in Fig. 2.1 at time $t_d = 0.0$ min.
Determine the gradient of this line, showing clearly how you did this.

gradient = $\text{cm}^3 \text{ min}^{-1}$ [2]

- (ii) Use your gradient to determine the rate of change of the amount of MnO_4^- ions required in mol min^{-1} .

rate of change of the amount of MnO_4^- ions required = mol min^{-1} [1]

- (iii) Determine the amount of H_2O_2 decomposed per minute and hence deduce the rate of change of $[\text{H}_2\text{O}_2]$ at $t_d = 0.0$ min, in $\text{mol dm}^{-3} \text{ min}^{-1}$.

amount of H_2O_2 decomposed = mol min^{-1}

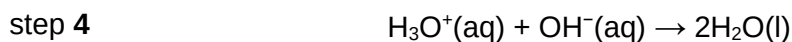
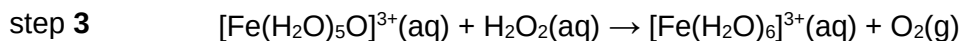
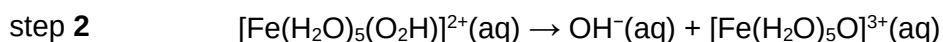
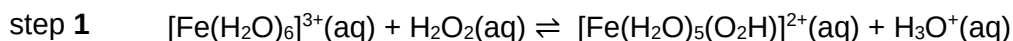
rate of change of $[\text{H}_2\text{O}_2]$ at $t_d = 0.0$ min = $\text{mol dm}^{-3} \text{ min}^{-1}$
[3]

- (iv) It has been claimed that the decomposition of hydrogen peroxide is first order with respect to $[\text{H}_2\text{O}_2]$.

State whether you agree or disagree with this claim. Use evidence from your graph in Fig. 2.1 to support your answer.

.....
.....
.....
..... [2]

- (c) Steps 1 to 4 represent a possible mechanism for the catalysed decomposition of H_2O_2 . In this mechanism, the O_2H ligand on one of the complex ions represents the $\text{H}-\text{O}-\text{O}^-$ ion and the O represents an oxygen atom.



- (i) Perform the test described in Table 2.1 and record your observations in the table. Test and identify any gases produced.

Table 2.1

test	observations
Put about 2 cm depth of H_2O_2 from the bench reagent bottle into a test-tube.	
Add about 2 cm depth of FA 4 to the same test-tube and shake the mixture thoroughly.	
Observe the mixture until no further changes are seen.	

[2]

- (ii) Identify from the mechanism in (c), the complex ions that are responsible for the colour changes you observed in Table 2.1.

.....

.....

..... [2]

- (iii) Explain fully how you can tell that the $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ ions are acting as a catalyst in this reaction, using evidence from your observations in Table 2.1 and from the mechanism.

.....

.....

.....

.....

.....

..... [3]

[Total: 24]

3 Qualitative Analysis

FA 8 is a solid oxide of an unknown metal **Z**. This oxide has an M_r **not** exceeding 140.

[A_r of Al: 27.0; Ba: 137.3; Ca: 40.1; Cr: 52.0; Cu: 63.5; Fe: 55.8; Mg: 24.3; Mn: 54.9; O: 16.0; Zn: 65.4]

FA 9 is solid potassium ethanedioate, $K_2C_2O_4$.

- (a) Perform the tests described in Table 3.1, and record your observations in the table. Test and identify any gases produced.

Table 3.1

test	observations
Add all of the solid FA 9 into a boiling tube. Add 15 cm ³ of FA 7 into this boiling tube. Gently warm the boiling tube and stir the mixture using the thermometer with 1 °C divisions , until the temperature of the mixture reaches about 60 °C. Turn off the Bunsen flame.	
Use a spatula to add FA 8 to the mixture, until no further change is seen. When you think that the reaction is complete, stop adding FA 8 . Filter the mixture into another boiling tube.	
To 1 cm depth of the filtrate in a test tube, add aqueous sodium hydroxide dropwise, until in excess.	

[4]

- (b) Explain how you have decided to stop adding **FA 8** at the point you have chosen.

.....

[1]

- (c) What is the chemical role of **FA 8** in the reaction between **FA 8** and **FA 9**? Support your answer using your observations from Table 3.1.

.....
.....
..... [1]

- (d) (i) Identify the cation in the filtrate, using your observations from Table 3.1.

.....
.....
..... [1]

- (ii) Hence, suggest the formula of the oxide of **Z** in **FA 8** and explain your reasoning by referring to your answer in (c) and the information provided on page 10.

.....
.....
.....
..... [2]

[Total: 9]

4 Planning

When some aqueous ethanoic acid is added to cyclohexane, the ethanoic acid distributes itself in the two immiscible solvents until an equilibrium is set up:



The equilibrium constant for this system is called the *partition coefficient*, K , which is a constant at a given temperature.

$$K = \frac{[\text{ethanoic acid}] \text{ in cyclohexane}}{[\text{ethanoic acid}] \text{ in water}}$$

To determine the value of K , it is necessary to determine the amount and then the concentration of ethanoic acid in each solvent layer at equilibrium. One way of doing this is described below.

Known volumes of aqueous ethanoic acid and cyclohexane are placed in a separating funnel, shown in Fig. 4.1, and mixed well.

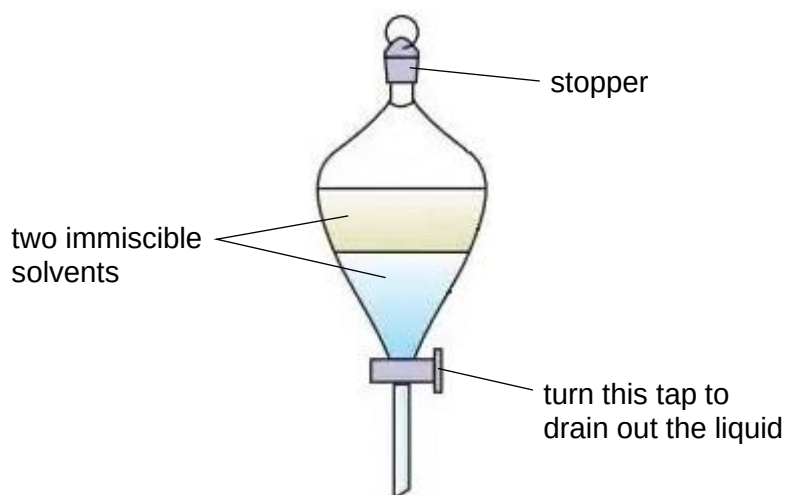


Fig. 4.1

Sufficient time must be provided to allow the following to occur.

- Distribution of the ethanoic acid molecules from the water solvent into the cyclohexane solvent until equilibrium is reached.
- Separation of the mixture into two immiscible aqueous and cyclohexane layers. (The density of cyclohexane is 0.78 g cm^{-3} . You may assume that the density of the aqueous layer is the same as that of water, 1.00 g cm^{-3} .)

Most of the aqueous layer is then taken out, from which a 10.0 cm^3 aliquot is withdrawn for titration.

The experiment is repeated using different volumes of aqueous ethanoic acid, and adding water such that the total volume of the aqueous and cyclohexane layers is kept constant.

Each titration only needs to be done once.

- (a) Using the information given, write a plan to collect sufficient data to plot a graph of the equilibrium concentration of ethanoic acid in cyclohexane against that in water, at room temperature.

You are provided with:

- 0.250 mol dm⁻³ aqueous ethanoic acid,
- cyclohexane,
- deionised water,
- 0.100 mol dm⁻³ aqueous sodium hydroxide,
- a separating funnel with capacity 250 cm³,
- other apparatus normally found in a school laboratory.

Your plan should include:

- essential details for the preparation of each equilibrium mixture,
- essential details for the titration of each aqueous aliquot against the sodium hydroxide using thymol blue indicator.

Thymol blue indicator has two pH working ranges as follows.

pH working range	colour in acidic region	colour in basic region
1.2 – 2.8	red	yellow
8.0 – 9.6	yellow	blue

Your first equilibrium mixture should be prepared using 50 cm³ of aqueous ethanoic acid and 50 cm³ of cyclohexane. You may draw a table to show the volumes of aqueous ethanoic acid, deionised water and cyclohexane used to prepare each equilibrium mixture.

You may assume that temperature stays constant throughout your experiment.

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the entire width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the paper.

- (b) Outline how you would calculate the concentration (in mol dm^{-3}) of ethanoic acid in each solvent, for **equilibrium mixture 1**, assuming that the titre volume is $V \text{ cm}^3$.

[4]

- (c) (i) Sketch on Fig. 4.2, a graph to show the relationship between the equilibrium concentrations of ethanoic acid in the two solvents.



Fig. 4.2

[1]

- (ii) How would you use this graph to determine a value for the partition coefficient, K , of ethanoic acid in cyclohexane and water?

..... [1]

- (d) An important assumption made in this experiment is that the solute *remains in the same molecular form* in the two solvents. Suggest how this assumption might **not** be true in this experiment.

.....

 [1]

[Total: 12]

Qualitative Analysis Notes

[ppt. = precipitate]

(a) Reactions of aqueous cations

cation	reaction with	
	NaOH(aq)	NH ₃ (aq)
aluminium, Al ³⁺ (aq)	white ppt. soluble in excess	white ppt. insoluble in excess
ammonium, NH ₄ ⁺ (aq)	ammonia produced on heating	–
barium, Ba ²⁺ (aq)	no ppt. (if reagents are pure)	no ppt.
calcium, Ca ²⁺ (aq)	white ppt. with high [Ca ²⁺ (aq)]	no ppt.
chromium(III), Cr ³⁺ (aq)	grey-green ppt. soluble in excess giving dark green solution	grey-green ppt. insoluble in excess
copper(II), Cu ²⁺ (aq)	pale blue ppt. insoluble in excess	blue ppt. soluble in excess giving dark blue solution
iron(II), Fe ²⁺ (aq)	green ppt., turning brown on contact with air insoluble in excess	green ppt., turning brown on contact with air insoluble in excess
iron(III), Fe ³⁺ (aq)	red-brown ppt. insoluble in excess	red-brown ppt. insoluble in excess
magnesium, Mg ²⁺ (aq)	white ppt. insoluble in excess	white ppt. insoluble in excess
manganese(II), Mn ²⁺ (aq)	off-white ppt., rapidly turning brown on contact with air insoluble in excess	off-white ppt. rapidly turning brown on contact with air insoluble in excess
zinc, Zn ²⁺ (aq)	white ppt. soluble in excess	white ppt. soluble in excess

(b) Reactions of aqueous anions

<i>anion</i>	<i>reaction</i>
carbonate, CO_3^{2-}	CO_2 liberated by dilute acids
chloride, $\text{Cl}^-(\text{aq})$	gives white ppt. with $\text{Ag}^+(\text{aq})$ (soluble in $\text{NH}_3(\text{aq})$)
bromide, $\text{Br}^-(\text{aq})$	gives pale cream ppt. with $\text{Ag}^+(\text{aq})$ (partially soluble in $\text{NH}_3(\text{aq})$)
iodide, $\text{I}^-(\text{aq})$	gives yellow ppt. with $\text{Ag}^+(\text{aq})$ (insoluble in $\text{NH}_3(\text{aq})$)
nitrate, $\text{NO}_3^-(\text{aq})$	NH_3 liberated on heating with $\text{OH}^-(\text{aq})$ and Al foil
nitrite, $\text{NO}_2^-(\text{aq})$	NH_3 liberated on heating with $\text{OH}^-(\text{aq})$ and Al foil; NO liberated by dilute acids (colourless $\text{NO} \rightarrow$ (pale) brown NO_2 in air)
sulfate, $\text{SO}_4^{2-}(\text{aq})$	gives white ppt. with $\text{Ba}^{2+}(\text{aq})$ (insoluble in excess dilute strong acids)
sulfite, $\text{SO}_3^{2-}(\text{aq})$	SO_2 liberated with dilute acids; gives white ppt. with $\text{Ba}^{2+}(\text{aq})$ (soluble in dilute strong acids)

(c) Test for gases

<i>gas</i>	<i>test and test result</i>
ammonia, NH_3	turns damp red litmus paper blue
carbon dioxide, CO_2	gives a white ppt. with limewater (ppt. dissolves with excess CO_2)
chlorine, Cl_2	bleaches damp litmus paper
hydrogen, H_2	"pops" with a lighted splint
oxygen, O_2	relights a glowing splint
sulfur dioxide, SO_2	turns aqueous acidified potassium manganate(VII) from purple to colourless

(d) Colour of halogens

<i>halogen</i>	<i>colour of element</i>	<i>colour in aqueous solution</i>	<i>colour in hexane</i>
chlorine, Cl_2	greenish yellow gas	pale yellow	pale yellow
bromine, Br_2	reddish brown gas / liquid	orange	orange-red
iodine, I_2	black solid / purple gas	brown	purple